

ORIGINAL PROBLEMS

Volume I · Trigonometry & Algebra

Four original problem constructions

“People wrote poems in my age.
I wrote problems in my pages.”

1. AP-GP-HP Trigonometric Structure

Problem

If $\cos \theta$, $\cos \phi$, $\sin \theta$ are in AP; $\cos \phi$, $\sin \theta$, $\tan \phi$ are in GP; and $\cos \theta$, $\sin \theta$, $\tan \phi$ are in HP, determine θ and ϕ and evaluate $\cos(\pi/3 + \theta) + \sin(\pi/6 + \phi)$.

Solution

Let $a = \cos \theta$, $b = \cos \phi$, $c = \sin \theta$ and $d = \tan \phi$. The conditions give $2b = a + c$, $c^2 = bd$, and $2/c = 1/a + 1/d$. Together with $a^2 + c^2 = 1$, the acute solution is $a = 2/\sqrt{5}$, $c = 1/\sqrt{5}$, $b = 3/(2\sqrt{5})$, $d = 2/(3\sqrt{5})$. Hence $\theta = \tan^{-1}(1/2)$, $\phi = \sin^{-1}(1/5)$, and the required value is $(7\sqrt{5} + 2\sqrt{3} - 2\sqrt{15})/20$.

2. Two-Angle Sine-Cosine Constraint

Problem

Given $\sin \theta + \sin \varphi = 1/8$, $\cos \theta + \cos \varphi = 1/4$, and $\theta > \varphi$, find $\tan(\pi/3 + \theta) + \cot(\pi/3 + \varphi)$.

Solution

Let $A = (\theta + \varphi)/2$ and $B = (\theta - \varphi)/2$. Then $2 \sin A \cos B = 1/8$ and $2 \cos A \cos B = 1/4$, so $\tan A = 1/2$. Also $2 \cos B = \sqrt{(1/8)^2 + (1/4)^2} = \sqrt{5}/8$, hence $\cos B = \sqrt{5}/16$. The ordering $\theta > \varphi$ fixes $B > 0$. Numerically, $\theta \approx 1.89423$ and $\varphi \approx -0.96694$ radians, giving $\tan(\pi/3 + \theta) + \cot(\pi/3 + \varphi) \approx 12.2295406285$.

3. The $\pi/9$ Polynomial Trap

Problem

Define $x = \cos(\pi/9) + \cos(4\pi/9) + \cos(8\pi/9)$. Evaluate $x^{2026} - 3x^{2024} - x^{2023}$.

Solution

The cosine sum equals $\sin 10^\circ$. By the triple-angle identity, $4 \sin^3 10^\circ - 3 \sin 10^\circ = -1$, so $x^3 - 3x - 1 = 0$. Therefore $x^{2026} - 3x^{2024} - x^{2023} = x^{2023}(x^3 - 3x - 1) = 0$.

4. Shifted Trigonometric Parameter

Problem

Suppose $\tan(\pi/6 + \theta) = \cot(\pi/6 + \theta)$ and $\cos(\pi/3 + \theta) \sin(\pi/6 + \varphi) = n$. Find $\cos(\pi/3 + \theta) + \cos(2\pi/3 + \theta) + \sin(\pi/6 + \varphi) + \cos(\pi/6 + \varphi)$, using the principal/acute branch.

Solution

From $\tan(\pi/6 + \theta) = \cot(\pi/6 + \theta)$, $\tan^2(\pi/6 + \theta) = 1$. On the principal branch, $\theta = \pi/12$. Thus $\cos(\pi/3 + \theta) = (\sqrt{6} - \sqrt{2})/4$ and $\cos(2\pi/3 + \theta) = -\sqrt{2}/2$. The parameter condition gives $\sin(\pi/6 + \varphi) = n(\sqrt{6} + \sqrt{2})$. On the acute branch, $\cos(\pi/6 + \varphi) = \sqrt{1 - (8 + 4\sqrt{3})n^2}$. Hence the required expression is $(\sqrt{6} - 3\sqrt{2})/4 + n(\sqrt{6} + \sqrt{2}) + \sqrt{1 - (8 + 4\sqrt{3})n^2}$.